

Kiran A. Vance

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Executive Summary

Senior Software Engineer with 9 years of experience architecting **high-throughput distributed systems** and **cloud-native infrastructure**. Expert in optimizing container orchestration and service mesh architectures to support 100M+ monthly active users. Proven leader in driving engineering excellence through automated governance and high-performance backend engineering.

Education

University of Illinois Urbana-Champaign | Urbana, IL

Bachelor of Science, Computer Science

2008 - 2012

Technical Expertise

- **Languages:** Go (Expert), Rust (Advanced), C++, TypeScript, Bash, SQL.
- **Systems:** Kubernetes (Operator Pattern), Service Mesh (Istio), gRPC, GraphQL, Redis.
- **Cloud & Platform:** GCP (GKE, Pub/Sub), Terraform, Helm, Spinnaker, Datadog.
- **Leadership:** Architecture Review Board, SLI/SLO Definition, Capacity Planning, Mentorship.

Professional Experience

Senior Software Engineer | Cloudflare — Austin, TX

March 2021 – Present

- Engineered a **Global Content Routing** engine in **Go**, optimizing edge-node selection and reducing cold-start latency for serverless functions by **55%**.
- Developed a custom **Kubernetes Controller** to automate the lifecycle of 10,000+ isolated namespaces, reducing manual SRE toil by **40 hours per week**.
- Refactored internal telemetry pipelines to use **Rust**, resulting in a **30% reduction** in memory overhead across the global edge network.
- Mentored 8 mid-level engineers and implemented a "Design Doc" culture that reduced architectural regressions by **20%**.

Backend Engineer III | Uber — San Francisco, CA

July 2017 – Feb 2021

- Architected a **Real-time Geospatial Indexing** service using **Go** and **Marketplace-scale Redis**, handling **1.2M concurrent driver updates**.
- Led the transition from a REST-based architecture to **gRPC and Protobuf**, improving cross-service communication efficiency by **4x**.
- Optimized large-scale data migrations between **NoSQL** clusters (Cassandra), ensuring zero downtime for mission-critical dispatch services.
- Implemented automated **Canary Analysis** using Spinnaker, reducing the "Time to Detect" (TTD) for deployment failures from minutes to seconds.

Projects

"Nebula" | Multi-Cloud Traffic Orchestrator

- Designed a cross-cloud traffic shifter that dynamically routes requests between AWS and GCP based on real-time spot-instance pricing and latency metrics.
- Implemented a **Zero-Trust** identity layer that validates mTLS certificates for all internal service-to-service calls.

Tech Stack: Go, Istio, Envoy, Terraform, Kubernetes.

"Chronos" | High-Precision Event Scheduler

- Built a distributed, persistent task scheduler capable of firing **100k events per second** with millisecond precision and guaranteed delivery.
- Integrated a **Kafka-backed** retry mechanism to handle transient downstream service failures without data loss.

Tech Stack: Rust, Kafka, PostgreSQL, Docker, GitHub Actions.